

APPLICATIONS COMBINED BAYESIAN ESTIMATES OF COEFFICIENT ALPHA

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Summary: The combined Bayesian estimation of more than one reliability coefficient is considered. The usual assumption of equicorrelated normal data is made. More than one sample, with possibly different variances, is available, and the combined Bayesian estimate is found.

1. Introduction

Coefficient Alpha (Cronbach, 1951) is used as a measure of reliability in psychometric tests. References to it can be found for instance in Kaplan and Saccuzza (1993), Hulin, Drasgow and Parsons (1983), Van Zyl, Neudecker and Nel (2000). Suppose that p normally distributed variables, each with variance σ^2 , are measured. Denote this vector of responses by $X' = (x_1, \dots, x_p)$, and let $E(X) = 0$, $cov(X) = \Sigma$. The assumption of equicorrelation is made. The covariance matrix is of the form,

$$\Sigma = \sigma^2 \begin{bmatrix} 1 & \rho & \dots & \rho \\ \rho & 1 & \dots & \rho \\ \vdots & \vdots & & \vdots \\ \rho & \rho & \dots & 1 \end{bmatrix}. \quad (1.1)$$

Coefficient Alpha is defined as

$$\begin{aligned}
 \alpha &= \frac{p}{p-1} \left(1 - \frac{\sum_{j=1}^p \text{Var}(X_j)}{\text{Total Variance}} \right) \\
 &= \frac{p}{p-1} \left(1 - \frac{\sum_{j=1}^p \text{Var}(X_j)}{\mathbf{1}'\Sigma\mathbf{1}} \right) \\
 &= \frac{p\rho}{1 + (p-1)\rho}. \tag{1.2}
 \end{aligned}$$

The total variance is the variance of the sum of the p variables. The vector $\mathbf{1}' = \underbrace{1, \dots, 1}_{p \text{ times}}$. The following relation will be of importance in this work:

$$1 - \alpha = \frac{1 - \rho}{1 + (p-1)\rho}. \tag{1.3}$$

The covariance matrix Σ has two distinct eigenvalues (e.g. Press, 1972) $\lambda_1 = \sigma^2(1 + (p-1)\rho)$ and $\lambda_2 = \sigma^2(1 - \rho)$, where λ_2 has multiplicity $p-1$. Note that $\Sigma > 0$ for $\rho > -\frac{1}{p-1}$.

Suppose a sample of size N of responses is available. Let $n = N - 1$ and $S = \frac{1}{n}A$ denote the unbiased estimates of Σ . Let r denote the combined maximum likelihood estimate of ρ (Muirhead, 1992, p. 114),

$$r = 2 \sum_{i < j} a_{ij} / ((p-1) \text{tr } A). \tag{1.4}$$

Coefficient Alpha or α is estimated as

$$\begin{aligned}\hat{\alpha} &= \frac{p}{p-1} \left(1 - \frac{tr A}{\mathbf{1}' A \mathbf{1}'} \right) \\ &= \frac{pr}{1 + (p-1)r},\end{aligned}\quad (1.5)$$

and

$$1 - \hat{\alpha} = \frac{1-r}{1 + (p-1)r}. \quad (1.6)$$

Viana (1995) derived theory for the combined Bayesian estimation of a correlation, where separate estimates are available. Variances may differ between samples. This theory is applied to find the combined Bayesian estimate of coefficient Alpha (Cronbach's Alpha) where estimates from independent sources are available. Viana (1995) expressed the likelihood in terms of λ_1 and λ_2 , which reduces to two independent parts.

Viana (1995) then derived the posterior density of ρ , assuming a uniform prior on ρ .

This results in a density of the form:

$$g(\rho|r) \propto \frac{\{1 + (p-1)\rho\}^{\frac{(p-1)n}{2}} (1-\rho)^{n/2}}{[\{1 + (p-1)\rho\} (p-1)(1-r) + (1-\rho)(1 + (p-1)r)]^{\frac{np}{2}}}. \quad (1.7)$$

The result can be extended to the case where k estimates, $r_1, \dots, r_k$ are available. The variance assumed in each case, need not be similar. Suppose there are k samples of sizes $N_1, \dots, N_k$, and $n = n_1 + \dots + n_k$, where $n_j = N_j - 1$, $j = 1, \dots, k$. It is assumed that estimates of the samples are

independent. The combined posterior of ρ is:

$$g(\rho | r_1, \dots, r_k) \propto \prod_{j=1}^k g(\rho | r_j), \quad (1.8)$$

where $g(\rho | r_j) = g_j(\rho)$ denotes the posterior of ρ for the j -th sample.

In section 2 the combined Bayesian estimate of α , where estimates $\hat{\alpha}_1, \dots, \hat{\alpha}_k$ are available, is derived.

2. Results

The posterior density (Viana, 1995) for the combined estimate of ρ is:

$$g(\rho | r_1, \dots, r_k) \propto \frac{\{1 + (p-1)\rho\}^{\frac{(p-1)n}{2}} (1-\rho)^{n/2}}{\prod_{j=1}^k [(1 + (p-1)\rho)(p-1)(1-r_j) + (1-\rho)(1 + (p-1)r_j)]^{\frac{n_j p}{2}}}. \quad (2.1)$$

The Jacobian of the transformation from ρ to α is

$$|J(p \rightarrow \alpha)| \propto \left(1 - \left(\frac{p-1}{p}\right)\alpha\right)^{-2}. \quad (2.2)$$

By using relations (1.2) and (1.6) it follows that the posterior of α is

$$f(\alpha | \hat{\alpha}_k) \propto \frac{(1-\alpha)^{n/2}}{\left(1 - \left(\frac{p-1}{p}\right)\alpha\right)^2 \prod_{j=1}^k \left(1 + \frac{1-\alpha}{(p-1)(1-\hat{\alpha}_j)}\right)^{\frac{n_j p}{2}}}. \quad (2.3)$$

The Jacobian above is also the transformation of a uniform prior on ρ to a prior on α defined on $(-\infty, 1)$. The prior density of α is:

$$f(\alpha) = \frac{p-1}{p^2} \left(1 - \left(\frac{p-1}{p}\right)\alpha\right)^{-2}. \quad (2.4)$$

The mean does not exist, but it can be shown that the median of f is

$$m = \frac{p-2}{p-1}. \quad (2.5)$$

This median is a monotone increasing function of p . It is well known in practice that coefficient Alpha tends to increase with p . The graph of $f(\alpha)$ is shown in figure 1.

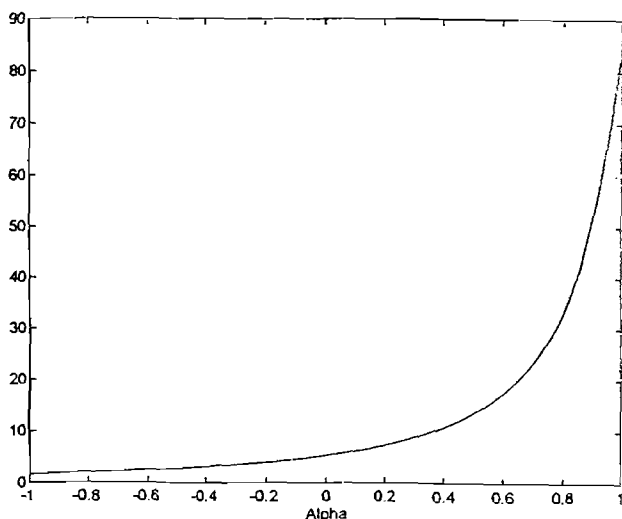


Figure 1. Prior density of α

Moments of the posteriors will be calculated numerically.

It can also be seen that by assuming a non-informative improper prior on α , of the form $f(\alpha) \propto 1$, will lead to a posterior without the Jacobian term in (2.3). The posterior of α in the combined case is

$$f(\alpha | \hat{\alpha}_1, \dots, \hat{\alpha}_k) \propto \frac{(1-\alpha)^{n/2}}{\prod_{j=1}^k \left(1 + \frac{(1-\alpha)}{(p-1)(1-\hat{\alpha}_j)} \right)^{\frac{n_j p}{2}}}. \quad (2.6)$$

3. Numerical examples

A questionnaire has been developed to measure the perceptions of customers of the service quality of a firm. One of the scales consists of four items, and is called access. Samples were drawn from three different locations. This resulted in three estimates of Alpha, $\hat{\alpha}_1 = 0.6605$ ($n_1 = 79$), $\hat{\alpha}_2 = 0.8139$ ($n_2 = 54$) and $\hat{\alpha}_3 = 0.7197$ ($n_3 = 45$).

The three correlation matrices are:

$$\text{Group 1: } \begin{bmatrix} 1.0000 & & & \\ 0.2969 & 1.0000 & & \\ 0.2893 & 0.4043 & 1.0000 & \\ 0.1611 & 0.4406 & 0.3769 & 1.0000 \end{bmatrix}$$

$$\text{Group 2: } \begin{bmatrix} 1.0000 & & & \\ 0.4170 & 1.0000 & & \\ 0.5422 & 0.6013 & 1.0000 & \\ 0.4229 & 0.6295 & 0.6130 & 1.0000 \end{bmatrix}$$

$$\text{Group 3: } \begin{bmatrix} 1.0000 & & & \\ 0.4505 & 1.0000 & & \\ 0.3416 & 0.5296 & 1.0000 & \\ 0.2177 & 0.3646 & 0.4516 & 1.0000 \end{bmatrix}$$

The graph of the posterior of the combined Bayesian estimate is shown in figure 2 and the three separate posterior densities in figure 3. In this case variances may have differed among samples, correlations were equal. A uniform prior on the correlation was assumed.

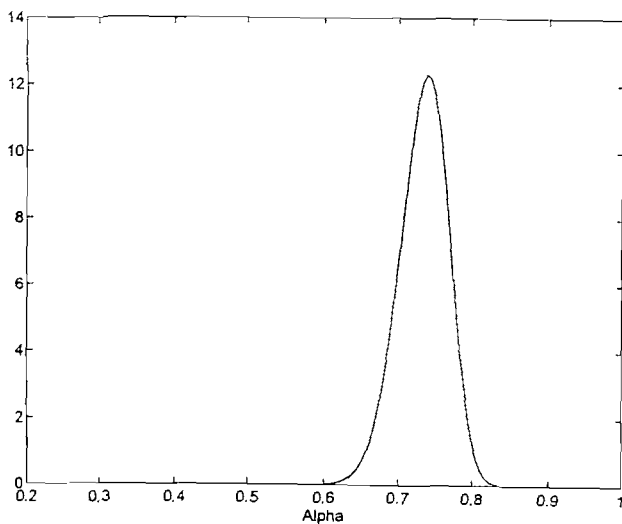


Figure 2. Posterior density of combined estimate of α

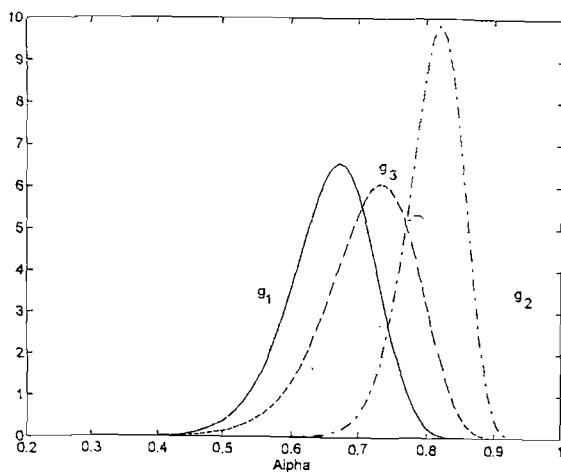


Figure 3. The individual posterior density of α_1 , α_2 and α_3

The graphs of the posteriors assuming a non-informative prior on Alpha is almost indistinguishable from those above and will not be shown here.

Moments were calculated by using numerical methods. The various estimates are shown in table 3.1. Estimated variances given in brackets.

Table 3.1. Summary of various estimates

	n	MLE estimate	Bayes Uniform prior on ρ $p(\rho) \propto 1$	Bayes Non-informative prior on α $p(\alpha) \propto 1$
α_1	79	0.6605	0.6577 (0.0039)	0.6459 (0.0042)
α_2	54	0.8139	0.8090 (0.0003)	0.8021 (0.0019)
α_3	45	0.7197	0.7141 (0.0048)	0.6983 (0.0050)
α	178	0.7273	0.7333 (0.0011)	0.7297 (0.0011)

The separate Bayesian estimates are slightly smaller than the MLE estimate, but the combined estimates are larger. In this example the estimates tend to be larger relative to the MLE of α , if the sample size is large.

4. Summary and discussion

Bayesian analysis gives a reasonably easy way to find the posterior of a combined α , where more than one estimate is available. Viana (1995) also considered the case where correlations are not assumed to be equal between groups. Simultaneous estimates for the correlations can then be found. The reasonable assumption of equal correlations is usually made, so this will not be considered further.

Winkler (1981), Lindley (1981) and various authors have discussed the combination of probability functions, also where dependence is assumed. Zellner and Palm (1992) derived theory for a predictive density for the combined estimate, where data is available from different sources. Theory along these lines can be exploited to find estimates which are correlated, or

to find approximations in complicated problems. If normality is assumed it often leads to a linear combination of the various estimates and t -distributions. Since exact distributions can be derived in the case of coefficient Alpha, this work will not be considered here.

References

- CRONBACH, L.J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika*, **6**, 297–334.
- HULIN, C.L., DRASGOW, F. & PARSONS, C.K. (1983). *Item response theory, application to psychological measurement*. Dorsey, Illinois.
- KAPLAN, R.M. & SACCUZZO, D.P. (1993). *Psychological testing, principles, applications and issues*. Brooks/Cole Publishing Company, Wadsworth, Belmont.
- LINDLEY D. (1983). Reconciliation of probability distributions. *Operations Research*, **31**, 866–880.
- MUIRHEAD, R.J. (1982). *Aspects of multivariate statistical theory*. John Wiley & Sons Inc., New York.
- PRESS, S.J. (1972). *Applied multivariate analysis*. Hold, Rinehart & Winston, New York.
- VAN ZYL, J.M., NEUDECKER, H. & NEL, D.G. (2000). On the distribution of the maximum likelihood estimator of Cronbach's Alpha. *Psychometrika*, **65**, 271–280.
- VIANA, M.A.G. (1995). Combined Bayesian estimates for the equicorrelation coefficient. *Canadian Journal of Statistics*, **23**, 45–53.
- WINKLER, R.L. (1981). Combining probability distributions from dependent information sources. *Management Science*, **27**, 479–488.
- ZELLNER, A. & PALM, F.C. (1992). To combine or not to combine? Issues of combining forecasts. *Journal of Forecasting*, **11**, 687–701.